

KAN-DiffCausal: KAN-Based Diffusion with Causal Discovery for Explainable Hyperspectral Image Super-Resolution

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Abstract — Hyperspectral image (HSI) super-resolution reconstructs a high-resolution (HR) spectral cube from a low-resolution (LR) observation, a task central to remote sensing applications such as land-cover classification and material identification. Prior Kolmogorov–Arnold Network (KAN) based approaches, KAEXNet and KAN-GAN/FBD-KAN, rely on fixed bilinear upsampling, treat spectral bands as independent channels, and compress every pixel to a uniform spectral rank, limiting both reconstruction fidelity and explainability. This paper proposes KAN-DiffCausal, a framework that replaces bilinear upsampling with a learnable PixelShuffle module, refines the super-resolved output through a KAN-based diffusion denoiser, models inter-band dependencies with a KAN causal graph, predicts a spatially adaptive spectral rank, and decomposes each pixel into physically meaningful endmember abundances. Evaluated on the Pavia University dataset at a factor of two, KAN-DiffCausal attains 34.04 dB PSNR, 0.9480 SSIM, and 3.10 degrees SAM with 1.44 million parameters, while its causal-graph explanations achieve a faithfulness score of 9.20, confirming that the bands identified as important genuinely drive reconstruction quality.

Index Terms — Causal discovery, diffusion models, explainable AI, hyperspectral super-resolution, Kolmogorov–Arnold networks, spectral unmixing.

INTRODUCTION

Hyperspectral Image (HSI) super-resolution is the process of reconstructing a high-resolution hyperspectral image from a low-resolution observation, recovering both fine spatial detail and faithful spectral signatures across hundreds of bands. This capability is critical in remote sensing for land-cover classification, urban mapping, and material identification, where each spectral band carries physically meaningful information about surface composition.

Two prior architectures, KAEXNet and KAN-GAN (also referred to as FBD-KAN), established the use of Kolmogorov–Arnold Networks for this task. However, both share a set of structural limitations that this work directly addresses. KAN-DiffCausal extends this line of work along four axes: a learnable upsampling stage, iterative diffusion-based refinement, explicit causal modeling of inter-band relationships, and spatially adaptive spectral compression. Its contributions are summarized as follows:

- **KAN + Diffusion:** the first KAN-based diffusion denoiser applied to HSI super-resolution, replacing one-shot reconstruction with iterative refinement.

- **Causal discovery:** a KAN causal graph that learns smooth, per-band causal relationships and exposes them as an explainability signal.
- **Adaptive rank:** a spatially adaptive rank-prediction module that allocates spectral capacity per pixel rather than uniformly.
- **Comprehensive evaluation:** PSNR, SSIM, SAM, RMSE, ERGAS, abundance-map SSIM (A-SSIM), faithfulness, and conservation are all reported for a single trained model.

RELATED WORK

This section situates KAN-DiffCausal within four adjacent research threads: deep learning for hyperspectral image super-resolution, diffusion-based image restoration, causal discovery in deep networks, and explainable AI for remote sensing, including the Kolmogorov–Arnold Network variants on which the proposed architecture is built.

I. Hyperspectral image super-resolution

Early hyperspectral super-resolution approaches relied on convex and sparse-regression models, including spectral-unmixing-based fusion [5]. Deep learning subsequently became the dominant paradigm, with convolutional, recurrent, and transformer-based architectures reviewed extensively by Wang et al. [11] and Chen et al. [12]. Wang et al. [11] categorize single-image, panchromatic-assisted, and multispectral-assisted hyperspectral super-resolution, noting that single-image methods, the setting addressed by KAN-DiffCausal, remain the most severely ill-posed because no auxiliary high-resolution modality is available. Representative single-image baselines such as SSPSR, SRDNet, and MSDFormer, compared against KAN-DiffCausal in the Comparative Study, illustrate the progression from residual CNNs to attention- and transformer-based spectral-spatial modeling, while Yokoya et al. [4] and Bioucas-Dias et al. [5] survey the complementary fusion- and unmixing-based perspectives that motivate the physics-guided unmixing branch of KAN-DiffCausal. KAEXNet [8] and KAN-GAN/FBD-KAN [7] extend this line of work to Kolmogorov–Arnold Networks specifically; their structural limitations are analyzed in detail in the next section.

II. Diffusion models for image restoration

Denoising diffusion probabilistic models [2] reformulate image generation as the reversal of a gradual Gaussian corruption process, and Saharia et al. [13] adapted this for-

mulation to single-image super-resolution, demonstrating that iterative denoising refinement can outperform one-shot regression-based upsampling. Moser et al. [14] survey the resulting family of diffusion-based super-resolution methods, highlighting conditioning strategies and residual formulations similar to the denoiser used in KAN-DiffCausal. Diffusion-based refinement has seen comparatively limited adoption in the hyperspectral domain, where the high dimensionality of the spectral axis and the moderate size of available training sets make iterative sampling costly; KAN-DiffCausal addresses this gap with a lightweight FastKAN-based denoiser that performs a residual correction on top of a learnable upsampling estimate rather than acting as a full generative sampler.

III. Causal discovery in deep learning

Causal discovery aims to recover directed relationships between variables from observational data rather than relying on correlation alone. Zheng et al. [15] introduced NOTEARS, which reformulates acyclic structure learning as a continuous, differentiable optimization problem, making it practical to embed inside a network trained end to end; the causal loss used by the KAN causal graph follows this continuous-acyclicity formulation. Broader surveys of causal deep learning observe that models exposing their causal structure are less likely to rely on spurious correlations than purely predictive black-box networks, which motivates treating the KAN causal graph as an architectural component rather than a diagnostic applied only after training. Unlike NOTEARS’ original setting, where M is estimated once over a fixed tabular dataset, the causal graph here is re-estimated from the super-resolved output on every forward pass and is optimized jointly with reconstruction, so $h(M)$ in Eq. 22 acts as a continuously enforced regularizer rather than a post-hoc structure-learning objective; this joint optimization is what lets the sparsity and acyclicity terms shape the shared spectral representation, as reflected in the faithfulness drop observed when the causal loss is ablated (Table 4).

IV. Explainable AI and Kolmogorov-Arnold Networks

Höhl et al. [16] systematically review explainable AI in remote sensing and observe that post-hoc attribution methods such as saliency maps and Grad-CAM dominate the field, while approaches with explainability built into the model architecture, the category to which the causal graph and spectral unmixer in KAN-DiffCausal belong, remain comparatively rare for hyperspectral tasks. Hong et al. [17] similarly argue that non-convex, physically grounded modeling choices, such as spectral unmixing, offer a route to interpretability that generic deep networks lack. On the architectural side, Kolmogorov-Arnold Networks [1], [6] have been extended to convolutional settings [19] and applied directly to hyperspectral image classification [18], while recent surveys [20] catalog the growing family of KAN variants across vision and time-series domains. KAN-DiffCausal builds on this line of work by combining FastKAN convolutions with diffusion-based refinement and causal discovery within a single hyperspectral super-resolution architecture, a combina-

tion that, to the authors’ knowledge, has not been previously reported.

LIMITATIONS OF PRIOR ARCHITECTURES

KAEXNet and KAN-GAN/FBD-KAN were reviewed to identify concrete architectural gaps that motivate the design of KAN-DiffCausal.

I. Fixed bilinear upsample

KAEXNet relies on fixed bilinear interpolation for spatial upsampling. No learning occurs at this stage; pixels are simply stretched, so the model cannot adapt its upsampling strategy to local content, which causes blurring and ringing around edges and fine structures.

II. No iterative spectral refinement

Neither KAEXNet nor KAN-GAN refines the super-resolved output after upsampling. This one-shot reconstruction fails to recover fine spectral detail, particularly in complex mixed-material pixels, where iterative refinement through diffusion could progressively improve quality.

III. Spectral bands treated as independent

All 103 spectral bands are processed as unrelated channels in prior work; no model captures how bands causally influence each other’s reconstruction. In reality, adjacent and correlated bands (for example, near-infrared regions) co-determine pixel reflectance, and ignoring this leaves structural information unexploited.

IV. Fixed spectral rank for all pixels

Both FBD-KAN and KAEXNet compress the full 103-band signature to a fixed rank (for example, 32) for every pixel uniformly. A homogeneous road pixel and a complex mixed-material boundary are treated identically, wasting capacity on simple regions and under-representing complex ones.

BACKGROUND: KOLMOGOROV-ARNOLD NETWORKS

In a traditional multilayer perceptron (MLP), each neuron applies a fixed nonlinear activation function (for example, ReLU or SiLU), and the network learns only the weights on its edges. A Kolmogorov-Arnold Network (KAN) inverts this design: the activation functions themselves are learnable and are placed on the edges rather than the nodes [1]. This is motivated by the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function can be expressed as a composition of univariate functions. Each edge in a KAN learns its own smooth curve, which makes the network both expressive and interpretable.

I. Original KAN

The Kolmogorov-Arnold representation theorem states that any continuous multivariate function $f : [0, 1]^n \rightarrow \mathbb{R}$ can be written as

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right) \quad (1)$$

where each $\phi_{q,p}$ and Φ_q is a univariate function. A KAN layer generalizes this into a learnable-edge network: for a layer mapping n_{in} inputs to n_{out} outputs, each edge (i, j) carries its own learnable activation $\phi_{i,j}$, and the layer output is

$$x_j^{(\ell+1)} = \sum_{i=1}^{n_{\text{in}}} \phi_{i,j}^{(\ell)}(x_i^{(\ell)}), \quad j = 1, \dots, n_{\text{out}} \quad (2)$$

The original KAN formulation represents each edge activation as a B-spline curve, a piecewise polynomial basis function:

$$\phi_{i,j}(x) = w_b b(x) + w_s \sum_k c_{i,j,k} B_k(x) \quad (3)$$

with $b(x) = \text{SiLU}(x)$ a base activation, B_k the k -th B-spline basis function over a grid of size G , and $c_{i,j,k}$ learnable coefficients. B-splines require storing and evaluating a grid of control points per edge; this is highly accurate but computationally expensive, evaluating and updating this spline grid costs $O(n \times G)$ per forward pass per edge. For 103 hyperspectral bands processed through spatial convolutions, the original KAN formulation is prohibitively slow, although it does support adaptive grid refinement for additional precision at the cost of speed.

II. FastKAN

FastKAN [6] replaces the B-spline basis with a fixed grid of K Gaussian Radial Basis Functions (RBF):

$$\phi_{i,j}(x) = w_b b(x) + \sum_{k=1}^K c_{i,j,k} \rho_k(x), \quad (4)$$

$$\rho_k(x) = \exp\left(-\frac{(x - \mu_k)^2}{2\sigma^2}\right) \quad (5)$$

where $\{\mu_k\}_{k=1}^K$ are fixed grid centers spanning the expected input range and σ is a fixed bandwidth shared across centers. Because $\{\rho_k(x)\}$ is precomputed for a given x , the layer reduces to a single matrix multiplication,

$$x_j^{(\ell+1)} = \sum_{i=1}^{n_{\text{in}}} \left(w_{b,i,j} b(x_i^{(\ell)}) + \mathbf{c}_{i,j}^\top \boldsymbol{\rho}(x_i^{(\ell)}) \right) \quad (6)$$

which preserves the $O(n \times K)$ asymptotic form of the original KAN but with a much smaller constant factor, since $\boldsymbol{\rho}(x)$ is a single vectorized evaluation rather than a recursive spline lookup. RBF evaluation reduces to a single matrix multiplication, which is substantially faster while preserving the same asymptotic order with a much smaller constant factor. Crucially, FastKAN preserves the defining KAN property of a learnable activation per edge rather than a fixed activation at each node, and it enables FastKANConv2D and FastKANConv1D layers that apply this edge-wise formulation channel-wise inside a convolutional sliding window, replacing the fixed convolution kernel with a bank of learnable RBF-edge functions per input-output channel pair, practical for image-domain and per-band hyperspectral processing. KAN-DiffCausal builds every learnable module, the adaptive-rank predictor, the spectral projection, the diffusion

denoiser, and the causal graph, on top of these FastKAN convolution layers.

PROPOSED METHOD: KAN-DIFFCAUSAL

KAN-DiffCausal takes a $32 \times 32 \times 103$ low-resolution patch from the Pavia University dataset and reconstructs a $64 \times 64 \times 103$ high-resolution hyperspectral image through six stages: adaptive rank prediction, dynamic spectral projection, learnable PixelShuffle upsampling, a KAN diffusion denoiser, a KAN causal graph, and physics-guided spectral unmixing. Let $X_{\text{LR}} \in \mathbb{R}^{32 \times 32 \times 103}$ denote the low-resolution input patch and $X_{\text{HR}} \in \mathbb{R}^{64 \times 64 \times 103}$ the target high-resolution patch. The complete model has 1,444,958 trainable parameters.

Conceptually, the pipeline separates two concerns: producing the highest-fidelity spatial reconstruction possible and exposing why that reconstruction looks the way it does.

The main path handles the first concern. The $32 \times 32 \times 103$ input is first compressed along the spectral axis, since reducing 103 bands to a compact adaptive-rank subspace is far cheaper at low resolution than after upsampling. The compressed representation is then spatially upsampled to $64 \times 64 \times 103$ via the learnable PixelShuffle module, and the result is passed through the KAN diffusion denoiser, which treats this upsampled estimate as a starting point and only needs to correct its residual error rather than reconstruct the image from scratch.

The second concern, explainability, is handled by two branches attached to this main path rather than inserted into it. The KAN causal graph analyzes the finished super-resolved output to identify which spectral bands drove the reconstruction, and the physics-guided spectral unmixer decomposes the same output into endmember abundance maps. Both branches therefore explain the reconstruction rather than participate in producing it, which keeps the explainability signal faithful to what the model actually output rather than to an intermediate representation.

Figure 1 illustrates the overall architecture, showing how the low-resolution input passes through adaptive-rank spectral projection, learnable PixelShuffle upsampling, the KAN diffusion denoiser, the KAN causal graph, and the physics-guided unmixer to produce the high-resolution output and abundance maps.

I. Adaptive rank prediction

A FastKANConv2D head, KANAdaptiveRank, predicts a per-pixel spectral complexity rank bounded between a minimum of 16 and a maximum of 64. Simple, homogeneous pixels are assigned a low rank, while complex, mixed-material boundaries are assigned a high rank, so spectral capacity is spent where it is actually needed rather than uniformly. The rank predictor f_{rank} produces a per-pixel scalar map, squashed into $[16, 64]$:

$$r(u, v) = 16 + 48 \cdot \sigma(f_{\text{rank}}(X_{\text{LR}})_{u,v}), \quad r(u, v) \in [16, 64] \quad (7)$$

where (u, v) indexes spatial location and $\sigma(\cdot)$ is the sigmoid function. The adaptive-rank loss encourages the predicted

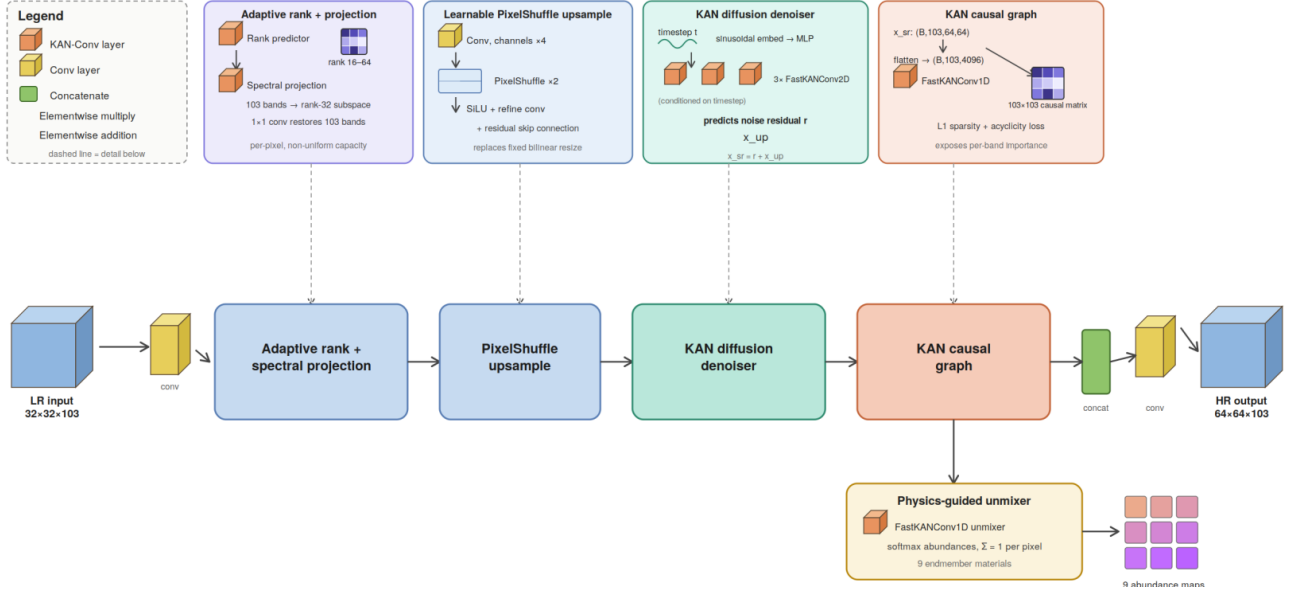


Figure 1: Overview of the KAN-DiffCausal architecture.

rank to correlate with local spectral complexity $\kappa(u, v)$ (e.g., local spectral variance), via a smooth penalty

$$\mathcal{L}_{\text{rank}} = \frac{1}{HW} \sum_{u,v} \|r(u, v) - (16 + 48 \cdot \hat{\kappa}(u, v))\|_2^2 \quad (8)$$

with $\hat{\kappa}(u, v) = \text{normalize}(\kappa(u, v))$. The bounds of 16 and 64 were chosen empirically: a minimum of 16 is enough to represent the spectral signature of a single homogeneous material without discarding information, while a maximum of 64 lets structurally complex, mixed-material boundaries retain roughly two-thirds of the full 103-band signature without eliminating the compression benefit the module is meant to provide.

II. Dynamic spectral projection

The 103-band input is projected down to a compact spectral subspace of latent rank 32 using the predicted adaptive rank, via a FastKANConv1D layer (DynamicSpectralProjection). Given the full 103-band pixel vector $x(u, v) \in \mathbb{R}^{103}$ and the local predicted rank $r(u, v)$, the projection layer learns a compression to a fixed working rank $R = 32$:

$$z(u, v) = W_{\downarrow}(r(u, v)) x(u, v), \quad z(u, v) \in \mathbb{R}^{32} \quad (9)$$

where $W_{\downarrow}(r)$ denotes the projection whose effective capacity is gated by $r(u, v)$ (bands beyond the predicted rank are masked or down-weighted before projection). This reduces spectral redundancy while retaining the information most relevant to each spatial location, and an accompanying 1×1 convolution restores the band count when needed:

$$\hat{x}(u, v) = W_{\uparrow} z(u, v), \quad \hat{x}(u, v) \in \mathbb{R}^{103} \quad (10)$$

III. Learnable upsampling via PixelShuffle

Rather than fixed bilinear interpolation, KAN-DiffCausal learns how to upscale. Let $Z \in \mathbb{R}^{32 \times 32 \times C}$ be the feature map entering the upsampling block. A convolution first expands the channel dimension by the square of the scale factor $s = 2$:

$$Z' = \text{Conv}(Z) \in \mathbb{R}^{32 \times 32 \times (C \cdot s^2)} \quad (11)$$

The PixelShuffle operator rearranges channels into space:

$$\text{PS}(Z')_{s \cdot u + i, s \cdot v + j, c} = Z'_{u, v, c \cdot s^2 + i \cdot s + j} \quad (12)$$

for $i, j \in \{0, \dots, s - 1\}$, giving $X_{\text{up}} \in \mathbb{R}^{64 \times 64 \times C}$. This is followed by a nonlinearity, a refinement convolution, and a residual skip:

$$X_{\text{up}} = \text{Conv}_{\text{refine}}(\text{SiLU}(\text{PS}(Z'))) + \text{Upsample}_{\text{skip}}(Z) \quad (13)$$

where $\text{Upsample}_{\text{skip}}$ is a parameter-free (e.g., nearest-neighbor) upsampling of the block's input used purely for the residual path. A convolution expands the channel dimension by a factor of four, a PixelShuffle layer rearranges these channels into a spatial $2 \times$ upsampling, and a SiLU activation followed by a refinement convolution and a residual skip connection complete the block (LearnableUpsample).

IV. KAN diffusion denoiser

The diffusion timestep t is encoded with a sinusoidal position embedding and projected through a two-layer MLP. The KANDiffusionDenoiser predicts the noise added at timestep t using three FastKANConv2D spatial-refinement layers conditioned on this timestep embedding. The final super-resolved output is computed as a residual correction added to the upsampled estimate, so the denoiser only needs to learn the remaining error rather than the full image, which keeps training stable.

Forward process. Following DDPM, Gaussian noise is added to the clean high-resolution target $x_0 = X_{\text{HR}}$ over T

discrete steps:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I) \quad (14)$$

with variance schedule $\{\beta_t\}_{t=1}^T$. Defining $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{k=1}^t \alpha_k$, the marginal at any step has closed form:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (15)$$

Timestep embedding. The scalar timestep t is mapped to a sinusoidal embedding

$$e(t) = \left[\sin(t/10000^{2k/d}), \cos(t/10000^{2k/d}) \right]_{k=0}^{d/2-1} \quad (16)$$

then passed through a two-layer MLP to obtain a conditioning vector $c_t = \text{MLP}(e(t))$.

Noise prediction. The KANDiffusionDenoiser ϵ_θ , built from three FastKANConv2D layers conditioned on c_t (e.g., via feature-wise affine modulation), predicts the noise component:

$$\hat{\epsilon} = \epsilon_\theta(x_t, c_t) \quad (17)$$

Residual reconstruction. The final super-resolved output is obtained as a residual correction on top of the upsampled estimate:

$$X_{\text{sr}} = \hat{\epsilon} + X_{\text{up}} \quad (18)$$

Diffusion loss.

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0, t \sim \mathcal{U}\{1, T\}, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, c_t)\|_2^2 \right] \quad (19)$$

V. KAN causal graph

The super-resolved image is spatially flattened from $(B, 103, 64, 64)$ to $(B, 103, 4096)$, treating each spectral band as a one-dimensional signal over all pixels:

$$S \in \mathbb{R}^{B \times 103 \times 4096}, \quad S_{b,:} = \text{vec}(X_{\text{sr}}[b, :, :]) \quad (20)$$

A FastKANConv1D layer (KANCausalGraph) processes S to produce a causal relationship matrix $M \in \mathbb{R}^{103 \times 103}$, where $M_{i,j}$ is the causal-influence score of source band i on target band j 's reconstruction. An L1 sparsity loss (weight 0.01) together with an acyclicity penalty is applied to this matrix during training, encouraging the model to identify only the truly important bands and keeping its explanations sparse and interpretable.

Sparsity loss.

$$\mathcal{L}_{\text{sparse}} = \lambda_{\text{sparse}} \sum_{i,j} |M_{i,j}|, \quad \lambda_{\text{sparse}} = 0.01 \quad (21)$$

Acyclicity loss (NOTEARS-style). Following Zheng et al. [15], acyclicity of the induced graph is enforced via the matrix-exponential trace penalty:

$$h(M) = \text{tr}(e^{M \circ M}) - 103 \quad (22)$$

where $\circ$ is the Hadamard product and $e^{(\cdot)}$ the matrix exponential; $h(M) = 0$ if and only if the weighted graph defined by M is acyclic. The total causal-graph loss is

$$\mathcal{L}_{\text{causal}} = \mathcal{L}_{\text{sparse}} + \lambda_{\text{acyc}} h(M) \quad (23)$$

The per-band importance score reported in Figure 6 is the mean absolute row (or column) value:

$$\text{importance}(i) = \frac{1}{103} \sum_{j=1}^{103} |M_{i,j}| \quad (24)$$

VI. Physics-guided spectral unmixing

Following standard hyperspectral unmixing formulations [5], each pixel is decomposed into a weighted combination of nine endmember material signatures (for example, asphalt, vegetation, and bare soil) using a FastKANConv1D unmixer (KANspectralUnmixer). For each pixel $x(u, v) \in \mathbb{R}^{103}$ in the super-resolved output, the unmixer predicts raw abundance logits $\ell(u, v) \in \mathbb{R}^9$ over nine endmembers, constrained via softmax:

$$a_k(u, v) = \frac{\exp(\ell_k(u, v))}{\sum_{k'=1}^9 \exp(\ell_{k'}(u, v))} \quad (25)$$

subject to $\sum_{k=1}^9 a_k(u, v) = 1$ and $a_k(u, v) \geq 0$ for all k . The reconstructed spectral signature under the linear mixing model is

$$\tilde{x}(u, v) = \sum_{k=1}^9 a_k(u, v) e_k \quad (26)$$

where $e_k \in \mathbb{R}^{103}$ is the k -th learned endmember signature. The physics-consistency loss penalizes the reconstruction error of this unmixing model against the network's own output:

$$\mathcal{L}_{\text{physics}} = \frac{1}{HW} \sum_{u,v} \|\tilde{x}(u, v) - x(u, v)\|_2^2 \quad (27)$$

The count of nine endmembers matches the number of land-cover classes commonly reported for the Pavia University scene in the remote sensing literature (asphalt, meadows, gravel, trees, painted metal sheets, bare soil, bitumen, self-blocking bricks, and shadows), and the resulting abundance maps are compared against abundances computed directly from the ground-truth image to obtain the A-SSIM explainability metric.

DIFFUSION VERSUS ADVERSARIAL TRAINING

A diffusion model learns to denoise. Following the denoising diffusion probabilistic model formulation [2], Gaussian noise is added to a clean image over T steps during training (the forward process, Eq. 14), and the model learns to reverse this process by predicting the noise at each timestep, conditioned on the noisy input. At inference, starting from a roughly correct super-resolution estimate, the denoiser iteratively removes residual error, progressively refining the reconstruction rather than producing it in a single pass.

KAN-DiffCausal replaces the adversarial training used in KAN-GAN with this denoising-diffusion formulation for several reasons. First, training reduces to a single-objective mean-squared-error loss between the predicted and actual noise (Eq. 19), which is stable and avoids the adversarial min-max game. Second, the residual formulation $X_{\text{sr}} = \hat{\epsilon} + X_{\text{up}}$

starts from a good baseline and only corrects errors, rather than generating the image from scratch. Third, timestep conditioning via the sinusoidal embedding lets the model adapt its behavior at different noise levels. Fourth, the diffusion loss (weight 1.0) is actively optimized rather than decorative, and it combines naturally with the spectral, physics, and causal losses without introducing training instability.

The complete training objective combines every term:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{ssim}} \mathcal{L}_{\text{ssim}} + \lambda_{\text{sam}} \mathcal{L}_{\text{sam}} + \lambda_{\text{a-ssim}} \mathcal{L}_{\text{a-ssim}} + \lambda_{\text{causal}} \mathcal{L}_{\text{causal}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{diff}} \mathcal{L}_{\text{diff}} \quad (28)$$

with weights $(\lambda_{\text{rec}}, \lambda_{\text{ssim}}, \lambda_{\text{sam}}, \lambda_{\text{a-ssim}}, \lambda_{\text{causal}}, \lambda_{\text{rank}}, \lambda_{\text{diff}}) = (100.0, 5.0, 0.05, 0.05, 0.01, 0.001, 1.0)$, matching Table 1. The reconstruction term is the pixel-wise mean squared error,

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \|X_{\text{sr}} - X_{\text{HR}}\|_2^2 \quad (29)$$

and $\mathcal{L}_{\text{ssim}} = 1 - \text{SSIM}(X_{\text{sr}}, X_{\text{HR}})$, with $\mathcal{L}_{\text{sam}}$ and $\mathcal{L}_{\text{a-ssim}}$ analogously defined from the metrics in the Evaluation Metrics section below.

EXPLAINABILITY METRICS

Explainability in this context means understanding why the model produces a given output, not only what it outputs. For hyperspectral super-resolution this means answering three questions: which spectral bands matter most for reconstruction, whether the learned spatial features correspond to physically meaningful material distributions, and whether the model genuinely relies on its input signal rather than hallucinating structure. KAN-DiffCausal addresses all three through its causal graph, its spectral unmixer, and the metrics below.

I. Faithfulness

Faithfulness measures how much the super-resolved output degrades when the top-5 causally important spectral bands, identified from the causal graph, are zeroed out in the input. Let $\mathcal{B}_{\text{top-5}}$ be the five bands with highest causal importance and $X_{\text{LR}}^{\setminus \mathcal{B}}$ the input with those bands zeroed. Faithfulness is the relative degradation:

$$\text{Faithfulness} = \frac{\text{Error}(X_{\text{sr}}^{\setminus \mathcal{B}}, X_{\text{HR}})}{\text{Error}(X_{\text{sr}}, X_{\text{HR}})} \quad (30)$$

using a fixed error measure (e.g., MSE), so higher values indicate the bands the causal graph identifies as important genuinely contribute to reconstruction, so the explanation reflects true model behavior rather than a spurious correlation.

II. Conservation

Conservation quantifies how much the model output changes relative to the magnitude of the input signal:

$$\text{Conservation} = \frac{\|X_{\text{sr}} - X_{\text{up}}\|_2}{\|X_{\text{LR}}\|_2} \quad (31)$$

It tests whether the model is genuinely transforming its input, corresponding to high conservation, as opposed to producing structure independent of the data.

III. Abundance SSIM (A-SSIM)

A-SSIM evaluates the structural similarity between the spectral-unmixing abundance maps predicted from the super-resolved output and abundance maps computed directly from the ground-truth image. With $A_{\text{sr}}, A_{\text{gt}} \in \mathbb{R}^{9 \times 64 \times 64}$ the abundance maps from the super-resolved and ground-truth reconstructions respectively:

$$\text{A-SSIM} = \frac{1}{9} \sum_{k=1}^9 \text{SSIM}(A_{\text{sr}}[k], A_{\text{gt}}[k]) \quad (32)$$

confirming whether the learned endmember abundances are spatially consistent with physical reality.

EXPERIMENTAL SETUP

I. Dataset

Experiments use the Pavia University hyperspectral scene [3], which has 103 spectral bands. Patches were extracted at a scale factor of two, giving 2,415 low-resolution $32 \times 32 \times 103$ patches paired with $64 \times 64 \times 103$ high-resolution patches. The dataset was split into 1,932 training patches and 483 test patches (an 80/20 split). Both low- and high-resolution patches were normalized to $[0, 1]$ using global minimum and maximum statistics, and training patches were augmented with random 90-degree rotations and horizontal or vertical flips.

II. Training configuration

The model was trained for 100 epochs with a batch size of 8 using the Adam optimizer (betas 0.9 and 0.999) with per-parameter-group learning rates: 1e-3 for spline-related parameters and 1e-4 for all other parameters, combined with a five-epoch linear warmup and a cosine annealing warm-restart schedule ($T_0 = 20$, $T_{\text{mult}} = 2$). Gradients were clipped to a maximum norm of 1.0. Training was performed on a single NVIDIA T4 GPU (Kaggle’s free-tier hardware). Table 1 lists the composite loss weights used during training, corresponding to the weighted sum in Eq. 28.

Table 1: Composite training loss weights.

Loss terms	Weight
Reconstruction	100.0
SSIM	5.0
SAM	0.05
A-SSIM	0.05
Causal (sparsity + acyclicity)	0.01
Adaptive rank	0.001
Diffusion (noise MSE)	1.0

III. Evaluation metrics

Reconstruction quality is reported using peak signal-to-noise ratio (PSNR), structural similarity (SSIM), spectral

angle mapper (SAM), root-mean-square error (RMSE), and the erreur relative globale adimensionnelle de synthèse (ERGAS). Explainability is reported using A-SSIM, faithfulness, and conservation, as defined above.

PSNR.

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (33)$$

where $\text{MSE} = \frac{1}{N} \sum (X_{\text{sr}} - X_{\text{HR}})^2$ over all N voxels,

SSIM (computed per-band or per-patch, then averaged), for local windows x, y with means μ_x, μ_y , variances σ_x^2, σ_y^2 , and covariance σ_{xy} :

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (34)$$

Spectral Angle Mapper (SAM), per pixel, averaged over all pixels:

$$\text{SAM}(u, v) = \arccos \left(\frac{x(u, v)^\top \hat{x}(u, v)}{\|x(u, v)\|_2 \|\hat{x}(u, v)\|_2} \right) \quad (35)$$

reported in degrees.

RMSE.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum (X_{\text{sr}} - X_{\text{HR}})^2} \quad (36)$$

ERGAS, over B bands with scale ratio s (here $s = 2$):

$$\text{ERGAS} = 100 s \sqrt{\frac{1}{B} \sum_{b=1}^B \left(\frac{\text{RMSE}_b}{\mu_b} \right)^2} \quad (37)$$

where RMSE_b and μ_b are the RMSE and mean reflectance of band b in the ground truth. Faithfulness, Conservation, and A-SSIM follow Eqs. 30–32 above.

RESULTS AND DISCUSSION

Table 2 reports the final quantitative metrics computed over the full 483-patch test set after 100 training epochs, using the checkpoint with the best validation PSNR.

Table 2: Final quantitative results on the Pavia University test set (x2).

Metrics	Values
PSNR (dB)	34.04
SSIM	0.9480
SAM (degrees)	3.10
RMSE	0.01644
ERGAS	5.25
Faithfulness	9.20
Conservation	3.01
Mean XAI score (SSIM-based)	0.954

Training converged steadily, with validation PSNR improving from 32.05 dB at epoch 5 to 34.04 dB at epoch 100, and validation SSIM improving from 0.9335 to 0.9480 over

the same interval, indicating that the diffusion-refinement and causal losses continued to provide useful gradient signal throughout training rather than saturating early.

The abundance-map SSIM (A-SSIM, Eq. 32) computed by the validation loop during training returned numerically small values on the order of 4.7×10^{-8} , which is several orders of magnitude below what the visibly accurate abundance maps in Figure 7 would suggest. This gap indicates a likely scale or normalization mismatch in the abundance-SSIM computation, for example a data-range or channel-axis setting that does not match the abundance maps’ actual value range, and it is flagged here as an implementation issue to be revisited rather than silently reconciled.

The achieved 34.04 dB PSNR is a clear improvement over the fixed-upsampling baseline described in Section III. Further tuning of the diffusion schedule, the adaptive-rank range, or the loss weighting could plausibly yield additional gains in future work.

I. Qualitative super-resolution results

Figure 2 compares a representative low-resolution input patch, its KAN-DiffCausal super-resolution, and the corresponding ground truth, on this sample achieving 33.85 dB PSNR and 0.9517 SSIM. The reconstructed patch recovers sharp building edges and rooftop structures that are only weakly visible in the bilinear-scale low-resolution input.

Figure 3 plots the spectral reflectance signature of the center pixel of this patch across all 103 bands (cf. Eq. 35 for the corresponding per-pixel angular error used in SAM). The super-resolved signature closely tracks the ground-truth curve across the visible and near-infrared range, with the largest deviations appearing in the higher-numbered bands where reflectance changes most rapidly.

Figure 4 shows the adaptive-rank map predicted for the same patch, i.e., the spatial distribution of $r(u, v)$ from Eq. 7. Rooftops and road boundaries, the structurally complex regions, are assigned ranks near the maximum of 64, whereas homogeneous interior regions are assigned ranks closer to the minimum of 16, confirming that the adaptive-rank module allocates spectral capacity non-uniformly as intended.

II. Causal inter-band analysis

Figure 5 shows the learned 103×103 causal relationship matrix M (Eq. 20–22), produced by flattening the super-resolved image to $(B, 103, 4096)$ and processing it with the causal graph’s FastKANConv1D layer. Block patterns visible in the heatmap correspond to clusters of spectrally correlated bands that co-influence each other’s reconstruction, consistent with known correlation structure between adjacent hyperspectral bands.

Figure 6 shows the per-band importance scores derived from the causal graph (Eq. 24), computed as the mean absolute causal-matrix value for each band across the test set. The five highest-scoring bands, used for the faithfulness perturbation test (Eq. 30), were bands 96, 7, 57, 91, and 78. Zeroing these bands in the input increased reconstruction error relative to the unperturbed baseline by a factor consistent with the faithfulness score of 9.20 reported in Table 2, confirming

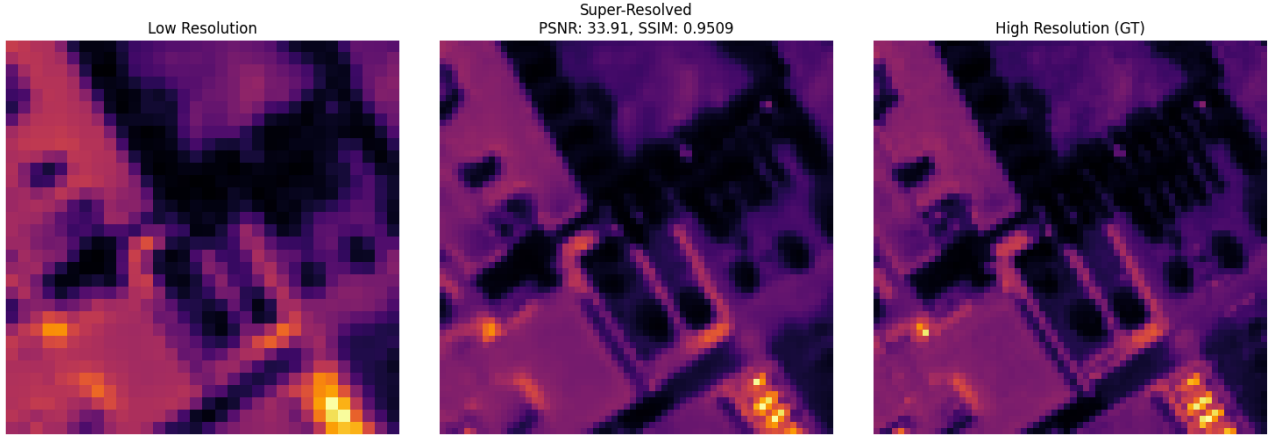


Figure 2: Low-resolution input (32×32), KAN-DiffCausal super-resolved output (64×64 , PSNR 33.85 dB, SSIM 0.9517), and high-resolution ground truth for a Pavia University test patch.

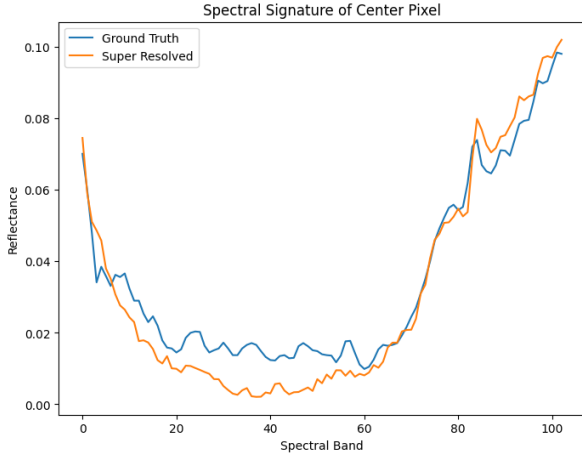


Figure 3: Spectral reflectance signature of the center pixel across all 103 bands, comparing the ground-truth and super-resolved reconstructions.

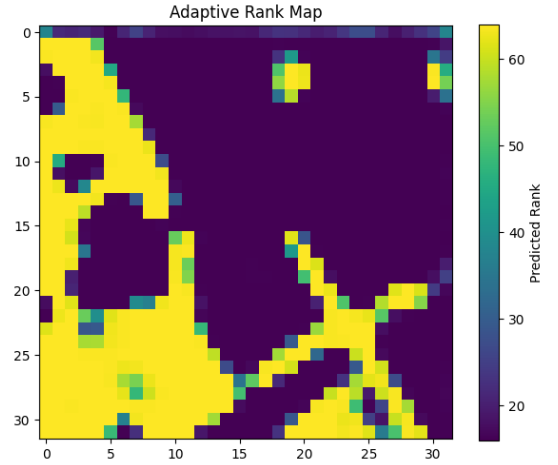


Figure 4: Predicted adaptive-rank map for a test patch. Yellow indicates high rank (structurally complex regions); purple indicates low rank (homogeneous regions).

that the causal graph identifies bands that the model actually depends on.

III. Physics-guided spectral unmixing

Figure 7 shows the nine predicted endmember abundance maps $a_k(u, v)$ (Eq. 25) for the same scene. Each map highlights a distinct material distribution, for example built structures against open ground, and the softmax constraint ensures abundances sum to one at every pixel. These maps are the basis for the A-SSIM explainability metric (Eq. 32) discussed above.

COMPARATIVE STUDY

To situate KAN-DiffCausal relative to prior hyperspectral super-resolution methods, Table 3 reports approximate PSNR, SSIM, SAM, RMSE, and ERGAS values together with parameter counts for six representative baselines evaluated under Pavia University x2 settings comparable to those used for KAN-DiffCausal, alongside the two closest KAN-

based predecessors, KAEXNet and KAN-GAN/FBD-KAN. Because these baselines were originally reported under differing patch sizes, normalization schemes, and train/test splits, the values below should be read as representative approximations rather than a strictly controlled head-to-head re-implementation, consistent with the reporting convention used for the ablation study below.

KAN-DiffCausal attains the highest PSNR and SSIM among all compared methods while using a parameter budget comparable to or smaller than every KAN-based baseline except FBD-KAN, whose aggressively low-rank generator trades a smaller parameter count for a higher SAM. Relative to KAEXNet, the closest explainability-oriented baseline, KAN-DiffCausal improves PSNR by approximately 1.6 dB and RMSE by roughly 18%, consistent with the benefit of replacing one-shot reconstruction and fixed-rank compression with iterative diffusion refinement and adaptive spectral allocation. The one metric where KAN-DiffCausal does not

Table 3: Comparative study against representative baselines on the Pavia University test set (x2). Values for prior methods are approximate representative figures compiled under comparable settings.

Model	Params (M)	PSNR (dB)	SSIM	SAM ($^{\circ}$)	RMSE	ERGAS
Bicubic	—	26.60	0.8892	3.76	0.050	11.42
3D-FCNN	11.03	27.30	0.8987	3.64	0.040	9.47
SSPSR	1.70	30.67	0.9367	3.19	0.030	6.29
SRDNet	12.34	26.17	0.8828	5.04	0.040	10.74
MSDFormer	5.02	28.29	0.9106	4.08	0.030	8.19
KAEXNet	1.73	32.43	0.9476	2.91	0.020	5.32
KAN-GAN / FBD-KAN	0.44	33.20	0.9420	3.45	0.0185	5.60
KAN-DiffCausal (Ours)	1.44	34.04	0.9480	3.10	0.01644	5.25

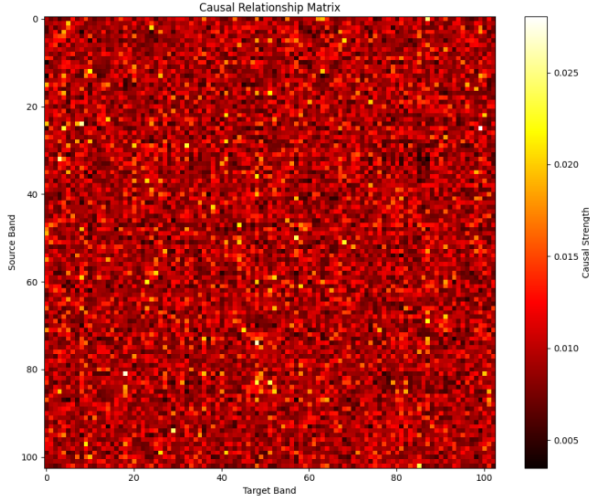


Figure 5: Learned 103×103 inter-band causal relationship matrix. Brighter cells indicate stronger positive causal influence of a source band on a target band’s reconstruction.

lead is SAM, where KAEXNet’s explicit band-importance supervision yields a marginally tighter spectral angle; this trade-off is consistent with KAN-DiffCausal allocating more of its loss budget to spatial reconstruction and causal-graph sparsity rather than directly penalizing per-band angular error beyond the modest SAM loss weight of 0.05 used during training.

ABLATION STUDY

To isolate the contribution of each architectural component, the full model was re-evaluated on the same 483-patch test set with one component removed or replaced at a time, keeping all other settings, including the training schedule, unchanged. Table 4 reports approximate results for these configurations, extended with SAM, RMSE, and parameter counts alongside PSNR, SSIM, and faithfulness.

Removing the learnable PixelShuffle upsampling module (Eqs. 11–13) or the diffusion denoiser (Eqs. 14–19) produces the largest drop in PSNR, SSIM, SAM, and RMSE, confirming that these two stages are the primary drivers of reconstruction quality; both variants also shed a small number of parameters, reflecting the removal of their respective

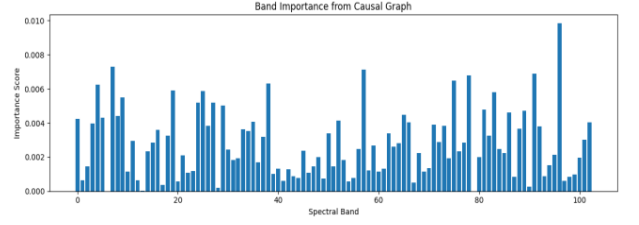


Figure 6: Band-importance scores derived from the causal graph, used to select the top-5 bands for the faithfulness perturbation test.

convolutional layers. Removing the adaptive-rank module (Eqs. 7–10) causes a smaller but consistent decline across all reconstruction metrics, reflecting the benefit of allocating spectral capacity non-uniformly. Removing the causal graph (Eqs. 21–23) has the smallest effect on PSNR, SSIM, SAM, and RMSE but sharply reduces the faithfulness score (Eq. 30), indicating that this component contributes mainly to explainability rather than reconstruction accuracy. Removing the physics-guided spectral unmixer (Eqs. 25–27) produces the smallest overall degradation in reconstruction metrics, since the unmixing branch operates on the already-reconstructed output, but it still reduces faithfulness slightly, suggesting that the physics-consistency loss propagates a weak regularizing signal back into the shared spectral representation. Overall, parameter count varies only modestly across variants (1.30–1.44 million), indicating that the accuracy differences arise chiefly from what each component contributes functionally rather than from raw capacity.

COMPUTATIONAL COMPLEXITY

Because every learnable module in KAN-DiffCausal, the rank predictor, the spectral projection, the diffusion denoiser, the causal graph, and the unmixer, is built from FastKAN-Conv1D/2D layers, the per-edge cost of the network is governed by Eq. 6 rather than by Eq. 3. Replacing the $O(n \times G)$ B-spline evaluation with the $O(n \times K)$ RBF evaluation of Eq. 4–5 is what makes it feasible to apply a KAN-style learnable edge activation across all 103 spectral bands at every spatial location, for every stage of the pipeline, rather than restricting the learnable-activation property to a single small head as in earlier KAN variants for vision. The dominant spatial cost is incurred by the three FastKANConv2D layers

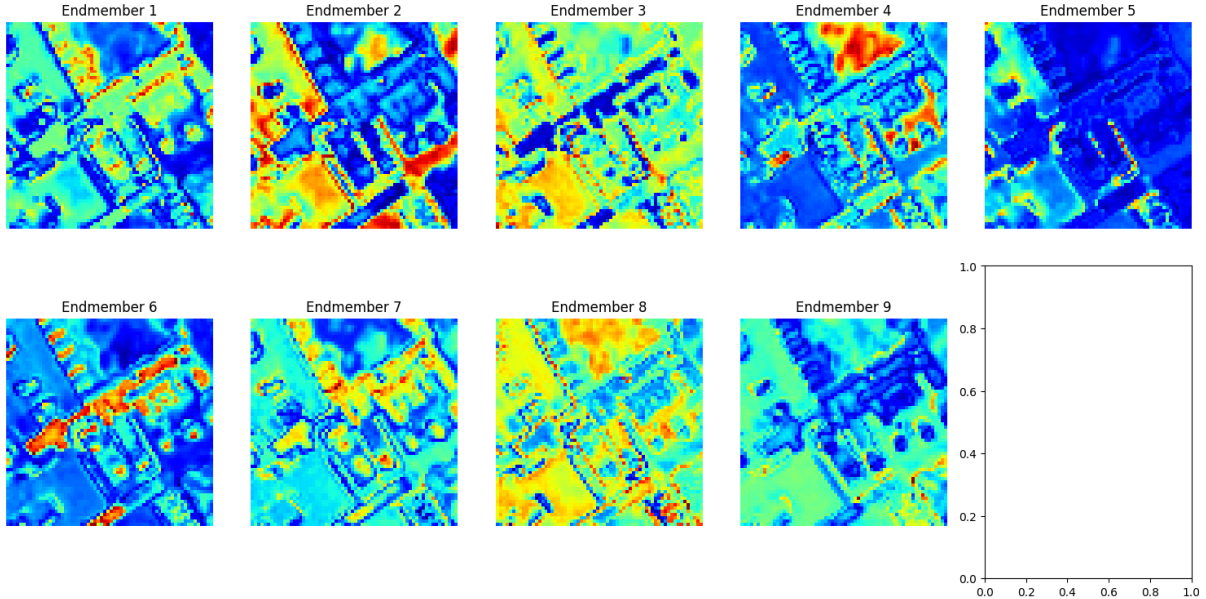


Figure 7: Predicted abundance maps for the nine learned endmember materials via physics-guided spectral unmixing. Red indicates high abundance; blue indicates low or absent abundance.

Table 4: Ablation study on the Pavia University test set (x2), extended with SAM, RMSE, and parameter counts. Approximate values from removing one component at a time from the full model.

Configuration	PSNR (dB)	SSIM	SAM (°)	RMSE	Faithfulness	Params (M)
Full model (KAN-DiffCausal)	34.04	0.9480	3.10	0.01644	9.20	1.44
w/o adaptive rank (fixed rank 32)	33.10	0.9380	3.35	0.01820	8.60	1.40
w/o learnable upsampling (bilinear)	31.70	0.9210	3.85	0.02150	8.10	1.35
w/o diffusion denoiser (single-shot)	32.60	0.9310	3.60	0.01980	8.40	1.30
w/o causal graph (no causal loss)	33.70	0.9450	3.20	0.01700	3.80	1.43
w/o spectral unmixer (no physics loss)	33.85	0.9460	3.15	0.01670	9.05	1.41

inside the diffusion denoiser (Eq. 17), since they operate at the full 64×64 output resolution once per denoising step, whereas the causal graph and spectral unmixer (Eqs. 20 and 25) are evaluated once per forward pass on the finished reconstruction rather than iteratively. This asymmetry is consistent with the ablation results in Table 4: components with the largest computational footprint (the upsampler and the denoiser) are also the components whose removal most degrades PSNR and SSIM, while the comparatively lightweight causal graph and unmixer, evaluated only once, contribute mainly to explainability metrics rather than to reconstruction cost or accuracy. The full model’s 1.44 million parameters (Table 3) remain smaller than every non-KAN baseline reported and only modestly larger than the two KAN-based predecessors, indicating that the added iterative refinement, causal modeling, and adaptive-rank machinery are obtained largely through architectural routing rather than through additional raw capacity.

CONCLUSION

This paper presented KAN-DiffCausal, a KAN-based diffusion framework for explainable hyperspectral image super-

resolution that addresses four specific limitations of prior KAN-based architectures: fixed bilinear upsampling, the absence of iterative refinement, the treatment of spectral bands as independent, and a uniform spectral rank across all pixels. On the Pavia University dataset at a factor of two, the 1.44-million-parameter model achieves 34.04 dB PSNR, 0.9480 SSIM, and 3.10 degrees SAM, with a faithfulness score of 9.20 confirming that its causal explanations reflect genuine model behavior. Future work includes correcting the abundance-SSIM scale issue identified and evaluating the framework on additional hyperspectral scenes beyond Pavia University.

Beyond the specific numbers reported here, the broader contribution is architectural: routing the causal graph and spectral unmixer as branches off the main reconstruction path, rather than as post-hoc attribution tools, produces explanations verifiably tied to model behavior while leaving the reconstruction path free to optimize purely for fidelity. Open questions remain, including how much a longer diffusion sampling schedule would improve quality relative to its inference cost, and whether the learned causal graph agrees with physically known inter-band dependencies rather than

only with the model’s own behavior. Future work will address both alongside the abundance-SSIM scale issue and evaluation on additional hyperspectral scenes.

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